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Real-Time Object Detection and using a Robotic Arm

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Setup

Raspberry Pi Mechanum Car and Desktop

- 2 pieces of hardware
- A raspberry pi controlled car with a servo arm
- A desktop PC with a dedicated GPU

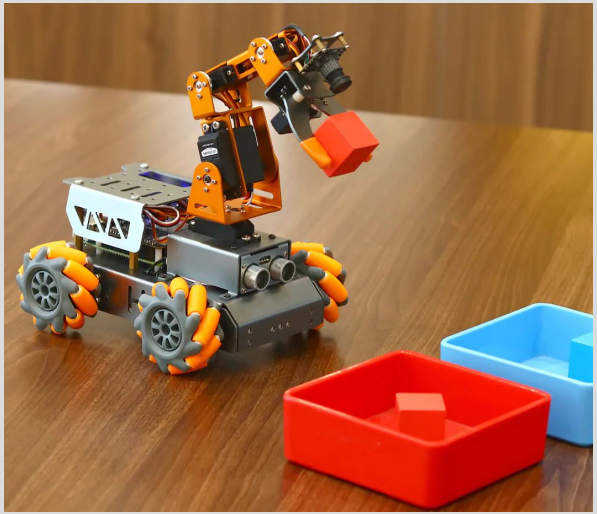
Streaming Communications Workflow

- Mjpeg server streams video feed to pc
- The desktop pc runs YOLO11n and computes next move
- Commands are sent to the car via a rpc server

Custom Made Dataset

- 255 images tailored to specific environment
- Annotated with label-studio

Features



Robust Object Detection

- With custom dataset achieved 98% map50
- Able to set 0.5 confidence to filter out misdetects

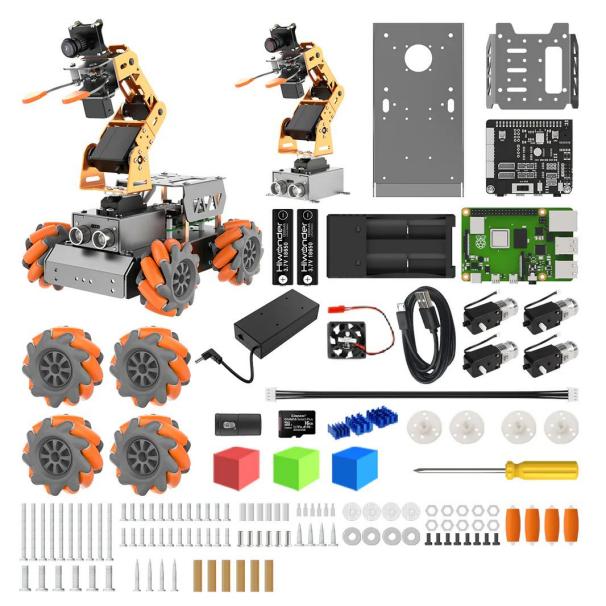
Mechanum Centering

- Converts image pixels to radians to predict adjustment requirement

Arm Stepping Algorithm

- Steps arm to minimize bounding box error
- Converts bounding box width to depth (1 camera)

Problems



Corrupted Dataset

- Originally trained on different environments. This caused misdetects
- Hsv_h automatic colour augmentation worsen results since cubes have the same shape.

Poor Camera

- Very easy for object to leave FOV when close
- Camera images blur when moving

Hard Problems

- How to know when an object is being held
- How to know when to close the claw to attempt pickup

Demo

https://youtube.com/shorts/V68eYEVw2bM?feature=share

Thank you